

Assignment: Digital Circuits (5-Hours)

1. Assignment-1 (Subjective-Problem Solving)

- 1.1. For each of the following circuits in Fig. 1, find the output and design a simpler circuit having the same output.

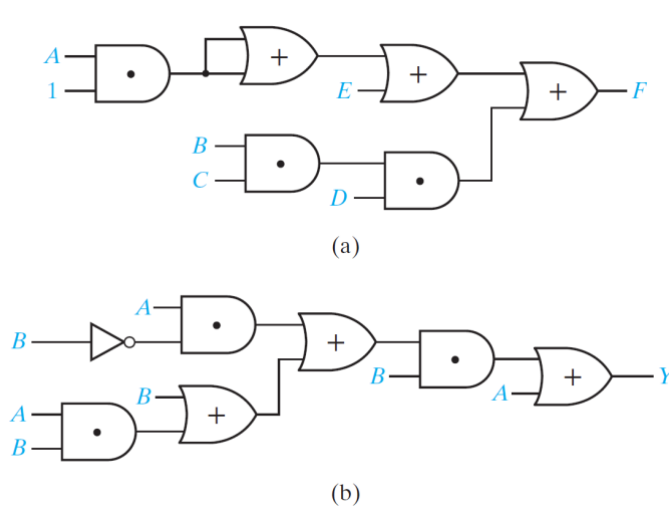


Fig. 1

- 1.2. Simplify each of the following expressions by applying one of the theorems.
- $X'YZ + (X'Y'Z)'$
 - $(AB' + CD)(B'E + CD)$
 - $(A'B + C + D)(A'B + D)$
- 1.3. For the following switching circuit in Fig. 2, find the logic function expression.

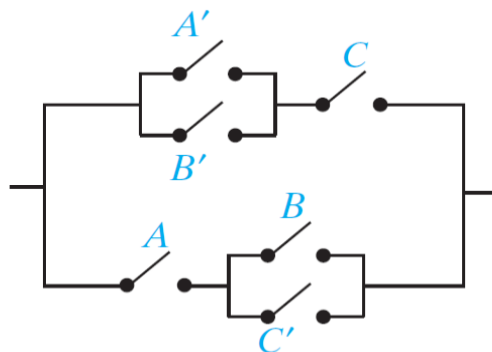


Fig.2

- 1.4. Show that the following two circuits in Fig. 3 realizes the same function.

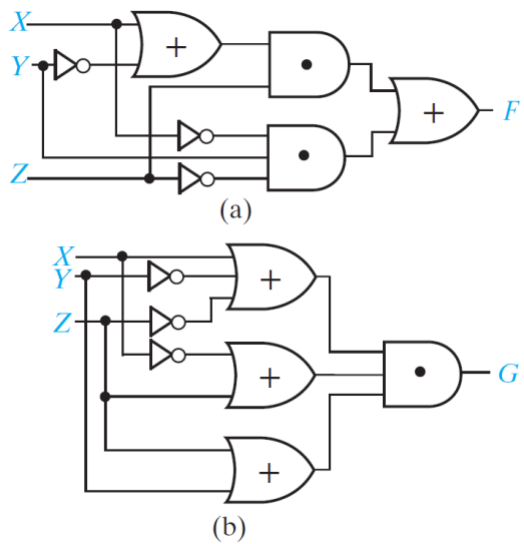


Fig.3

- 1.5. The following circuit is implemented using two half-adder circuits. The expressions for the half-adder outputs are $S = A \oplus B$, and $C = AB$. Derive simplified sum-of-products expressions for the circuit outputs SUM and C_o . Give the truth table for the outputs.

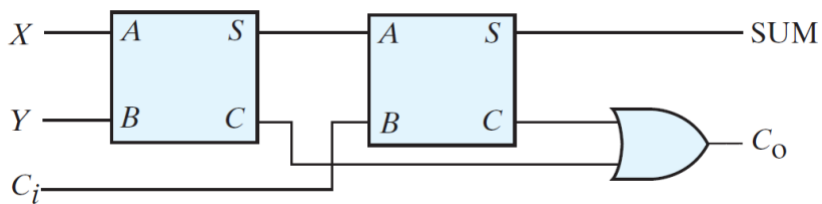


Fig.4

- 1.6. Convert the following circuit in Fig. 5 to all NAND gates, by adding bubbles and inverters where necessary.

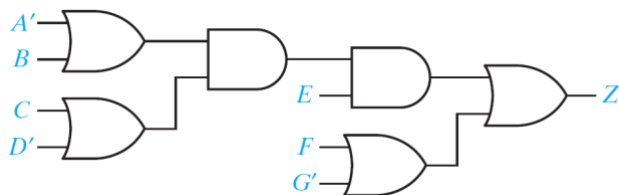


Fig. 5

- 1.7. Convert the Use gate equivalences to convert the circuit in Fig. 6 into a four-level circuit containing only NAND gates and a minimum number of inverters. (Assume the inputs are available only in uncomplemented form.)

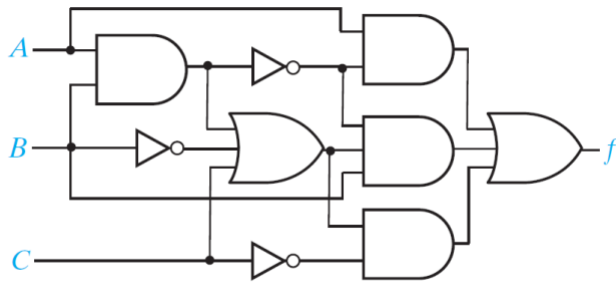


Fig. 6

- 1.8. A combinational switching circuit has four inputs and one output as shown in Fig. 7. $F = 0$ if three or four of the inputs are 1. Write the maxterm expansion for F .

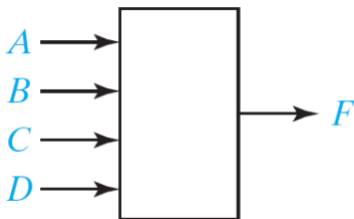


Fig. 7

- 1.9. Complete the timing diagram for the given circuit in Fig. 8. Assume that both gates have a propagation delay of 5 ns.

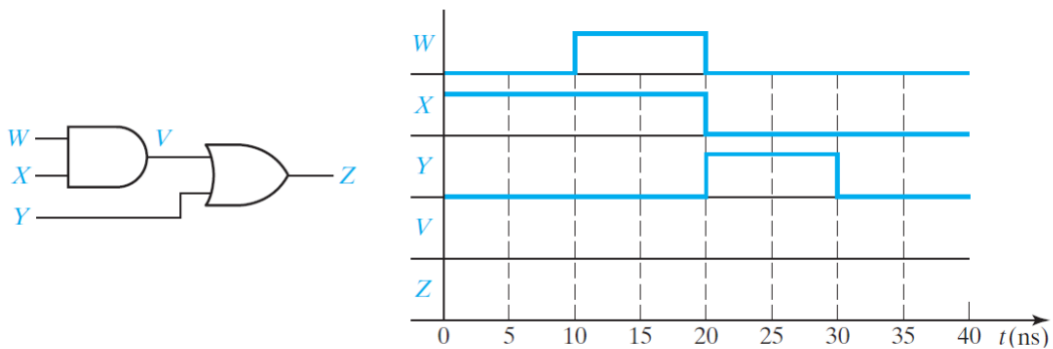


Fig. 8

- 1.10. For the following circuit in Fig. 9, assume that the inverters have a delay of 1 ns and the other gates have a delay of 2 ns. Initially $A = 0$ and $B = C = D = 1$, and C changes to 0 at time = 2 ns. Draw a timing diagram and identify the transient that occurs.

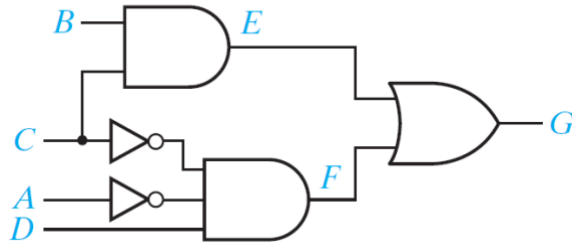


Fig. 9

- 1.11. Draw the timing diagram for V and Z for the circuit in the Fig. 10. Assume that the AND gate has a delay of 10 ns and the OR gate has a delay of 5 ns.

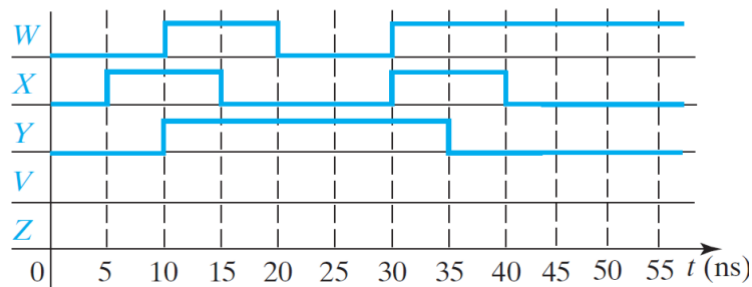
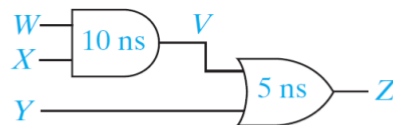


Fig. 10

- 1.12. The following PLA as shown in Fig. 11 will be used to implement the following equations. Indicate the connections that will be made to program the PLA to implement these equations.

$$X = AB'D + A'C' + BC + C'D'$$

$$Y = A'C' + AC + C'D'$$

$$Z = CD + A'C' + AB'D$$

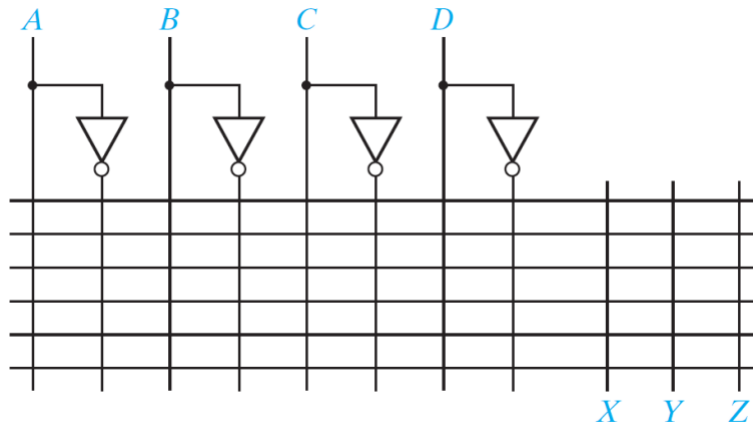


Fig. 11

- 1.13. Use a 4-to-1 multiplexer and a minimum number of external gates to realize the function $F(w, x, y, z) = \sum m(3, 4, 5, 7, 10, 14) + \sum d(1, 6, 15)$. The inputs are only available uncomplemented.
- 1.14. Show how to make an 8-to-1 MUX using two 4-to-1 MUXes, two three-state buffers, and one inverter.
- 1.15. In Fig. 12, the seven-segment indicator can be used to display any one of the decimal digits 0 through 9. For example, “1” is displayed by lighting segments 2 and 3, “2” by lighting segments 1, 2, 7, 5, and 4, and “8” by lighting all seven segments. A segment is lighted when logic 1 is applied to the corresponding input on the display module.

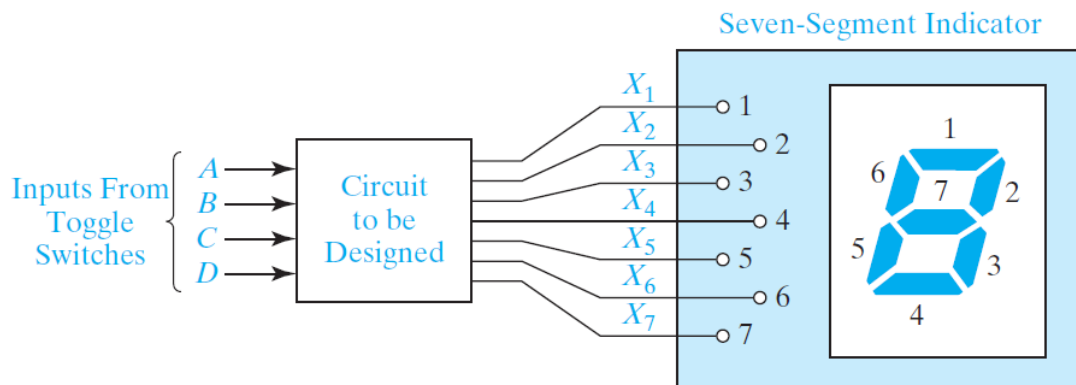


Fig. 12

- 1.16. Design a circuit which will yield the product of two binary numbers, n_2 and m_2 , where $00_2 \leq n_2 \leq 11_2$ and $000_2 \leq m_2 \leq 101_2$. For example, if $n_2 = 10_2$ and $m_2 = 001_2$, then the product is $n_2 \times m_2 = 10_2 \times 001_2 = 0010_2$. Let the variables A and B represent the first and second digits of n_2 , respectively (i.e., in this example $A = 1$ and $B = 0$). Let the variables C , D , and E represent the first, second, and third digits of m_2 , respectively (in this example $C = 0$, $D = 0$, and $E = 1$). Also let the variables W , X , Y , and Z represent the first, second, third, and fourth digits of the product. (In this example $W = 0$, $X = 0$, $Y = 1$, and $Z = 0$.) Assume that $m_2 > 101_2$ never occurs as a circuit input.

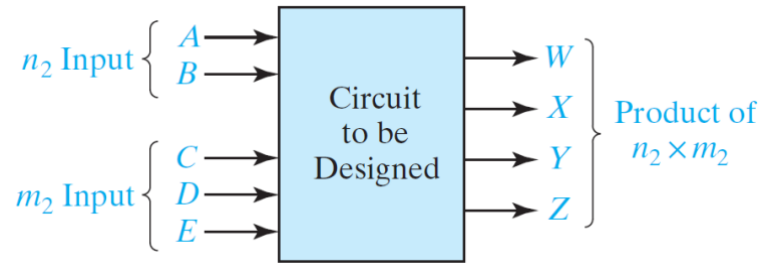


Fig. 13

- Design the circuit using only two-, three-, and four-input NOR gates and inverters. Try to minimize the total number of gates and inverters required. The variables A , B , C , D , and E will be available from toggle switches. Any solution that uses 15 or fewer gates and inverters (not counting the five inverters for the inputs) is acceptable.
- 1.17. Braille is a system which allows a blind person to read alphanumeric by feeling a pattern of raised dots. Design a circuit that converts BCD to Braille. The table 1 shows the correspondence between BCD and Braille.

				$\begin{array}{c c} W & X \\ \hline Z & Y \end{array}$	
A	B	C	D		
0	0	0	0	.	:
0	0	0	1	.	
0	0	1	0	:	
0	0	1	1	.	.
0	1	0	0	.	:
0	1	0	1	.	.
0	1	1	0	:	.
0	1	1	1	:	:
1	0	0	0	.	.
1	0	0	1	.	.

Table 1

- 1.18. A simple security system for two doors consists of a card reader and a keypad in the circuit in Fig. 13.

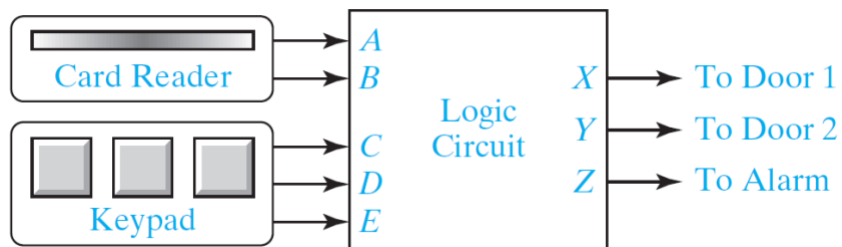


Fig. 13

A person may open a particular door if he or she has a card containing the corresponding code and enters an authorized keypad code for that card. The outputs from the card reader are as follows:

	A	B
No Card Inserted	0	0
Valid Code for Door 1	0	1
Valid Code for Door 2	1	0
Invalid Card Code	1	1

To unlock a door, a person must hold down the proper keys on the keypad and, then, insert the card in the reader. The authorized keypad codes for door 1 are 101 and 110, and the authorized keypad codes for door 2 are 101 and 011. If the card has an invalid code or if the wrong keypad code is entered, the alarm will ring when the card is inserted. If the correct keypad code is entered, the corresponding door will be unlocked when the card is inserted.

Design the logic circuit for this simple security system. Your circuit's inputs will consist of a card code AB , and a keypad code CDE . The circuit will have three outputs XYZ (if X or $Y = 1$, door 1 or 2 will be opened; if $Z = 1$, the alarm will sound). Design your circuit using only two-, three-, and four-input NOR gates and inverters. Any solution with 19 or fewer gates and inverters (not counting the five inverters for the inputs) is acceptable. Use toggle switches for inputs A , B , C , D , and E when you test your circuit.

- 1.19. Design a "disk spinning" animation circuit for a CD player. The input to the circuit will be a 3-bit binary number $A_1 A_2 A_3$ provided by another circuit. It will count from 0 to 7 in binary, and then it will repeat.). The animation will appear on the top four lights of the LED display of Figure 8-15, i.e., on X_1 , X_2 , X_7 , and X_6 , going clockwise. The animation should consist of a blank spot on a disk spinning around once, beginning with X_1 . Then, the entire disk should blink on and off twice. The pattern is shown in Fig. 14.



Fig. 14

Design your circuit using only two-, three-, and four-input NOR gates and inverters. Try to minimize the number of gates required. Any solution which uses 11 or fewer gates (not counting the four inverters for the inputs) is acceptable.

- 1.20. The circuit in Fig. 15 has a 4-input priority encoder connected to a 2-to-4 decoder with enable. The truth table for the priority encoder is given. (The I_3 input is highest priority.) All signals are active high. What functions of A , B , C , and D are realized by Z_3 , Z_2 , Z_1 , and Z_0 ?

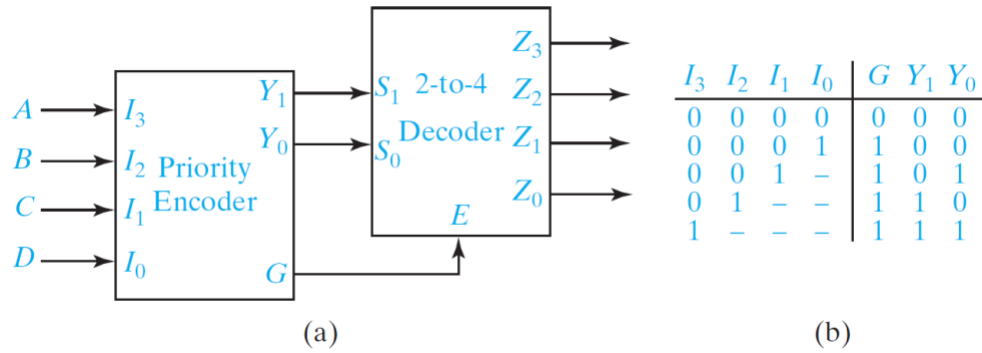


Fig. 15

- 1.21. The circuit in Fig. 16 has a 2-to-4 decoder with active high outputs connected to a 4-to-1 MUX with an active low output. Derive a minimum SOP or a minimum POS expression for the output, $f(A, B, C, D)$.

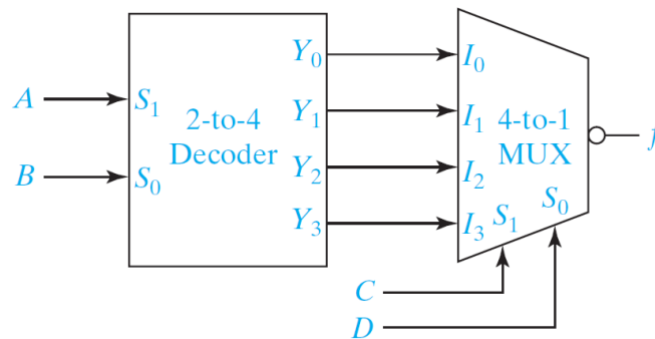


Fig. 16

- 1.22. Assume that the inverter in the circuit Fig. 17 has a propagation delay of 5 ns and the AND gate has a propagation delay of 10 ns. Draw a timing diagram for the circuit showing X , Y , and Z . Assume that X is initially 0, Y is initially 1, after 10 ns X becomes 1 for 80 ns, and then X is 0 again.

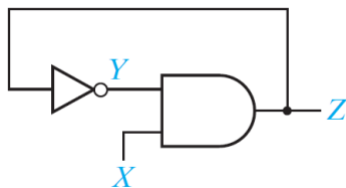


Fig. 17

- 1.23. A latch can be constructed from an OR gate, an AND gate, and an inverter connected as follows:

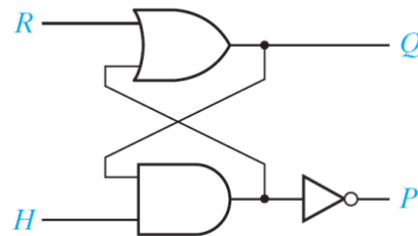
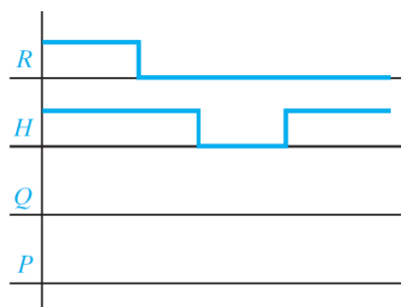


Fig. 17

- What restriction must be placed on R and H so that P will always equal Q' (under steady-state conditions)?
- Construct a next-state table and derive the characteristic (next-state) equation for the latch.
- Complete the following timing diagram for the latch.



- 1.24. This problem illustrates the improper operation that can occur if both inputs to an S-R latch are 1 and are then changed back to 0.

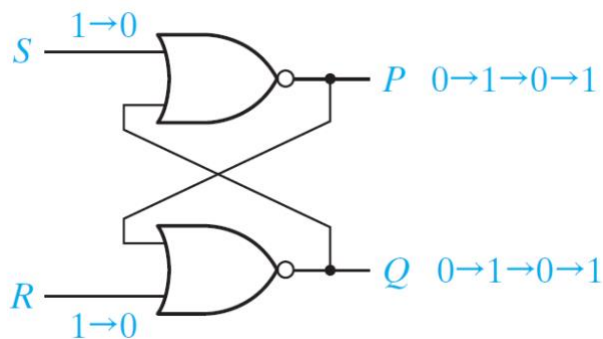
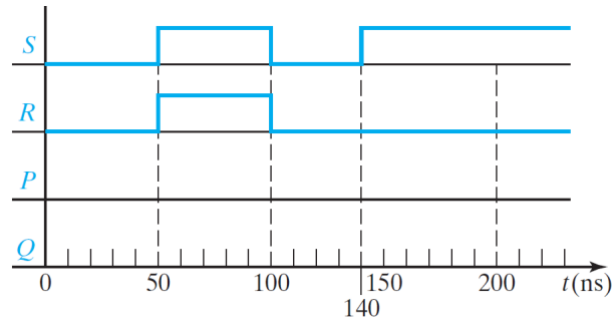


Fig. 18

For Fig. 18, complete the following timing chart, assuming that each gate has a propagation delay of exactly 10 ns. Assume that initially $P = 1$ and $Q = 0$. Note that when $t = 100$ ns, S and R are both changed to 0. Then, 10 ns later, both P and Q will change to 1. Because these 1's are fed back to the gate inputs, what will happen after another 10 ns?



- 1.25. What change must be made to Fig.19 to implement a falling-edge-triggered D flip-flop? Complete the following timing diagram for the modified flip-flop.

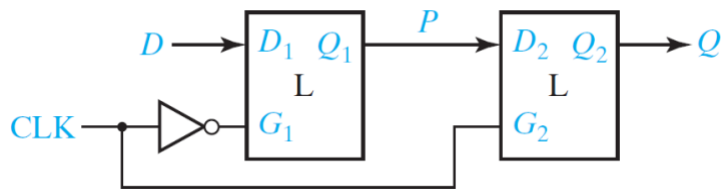
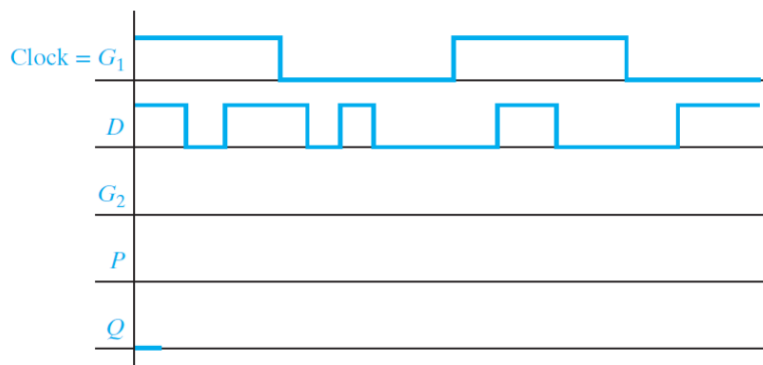


Fig. 19



- 1.26. Design a 3-bit counter which counts in the sequence: 001, 011, 010, 110, 111, 101, 100, (repeat) 001. Use D flip-flops.